resipsc v2.0.0

Regulatory Package — Residual PSC Detection Software

FDA · EMA · ICH · GAMP 5 · 21 CFR Part 11 · GMLP · PCCP

Document	resipsc Regulatory Package (RUO)
Software version	2.0.0 (resipsc-ensemble-2.0.0)
Model card	15850530e1aea67e
Intended use	Computational residual PSC detection in scRNA-seq of PSC-derived cell therapy products
Regulatory status	Research Use Only — NOT FDA-cleared
Audit trail	db/audit.db (SHA-256 hash-chained, append-only)

1. FDA Position and Applicable Regulations

1.1 21 CFR 1271 — Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps)

PSC-derived cell therapy products that are more than minimally manipulated fall under 21 CFR 1271 and 21 CFR 211 (cGMP). Release testing for residual undifferentiated PSCs is required before administration because transplanted residual PSCs can form teratomas. resipsc computes a per-batch residual PSC fraction and emits a disposition recommendation aligned with these rules: RELEASE at an observed fraction at or below 0.001% (with orthogonal confirmation), HOLD between 0.001% and 1% pending QA review, and QUARANTINE above 1%. The tool documents the biological rationale (pluripotency programme detection via OCT4/POU5F1, NANOG, SOX2 and an extended 8-marker panel) and maps detected cells to germ-layer lineages and 22 teratoma-relevant anatomical sites with literature-derived priors.

1.2 21 CFR Part 11 — Electronic Records and Electronic Signatures

The audit trail is implemented as an append-only SQLite event log in which every event carries a SHA-256 hash computed over the previous event's hash, the event payload and a UTC timestamp, forming a tamper-evident chain. No UPDATE or DELETE code path exists for audit rows; the chain can be verified programmatically via the /api/audit endpoint, and exports to CSV are provided for inspectors. The log captures user ID, timestamp, action, entity, old and new values, reason for change, and IP address. Records are retained for at least 5 years after batch release (10 years recommended for cell therapies). Electronic signature blocks are present on QC reports for QA/QP wet-ink or 21 CFR Part 11-compliant e-signature workflows.

1.3 Good Machine Learning Practice (GMLP)

#	GMLP principle (FDA/Health Canada/MHRA, Oct 2021)	resipsc implementation
1	Multidisciplinary expertise	ML, QA, bioinformatics and regulatory roles defined in FMEA team and RBAC
2	Good software engineering and security practices	Typed codebase, pinned dependencies (SBOM), Docker build, RBAC, session timeout
3	Data management reflecting intended use	Training corpus registry with provenance; data hash in model card
4	Data quality and relevance	Cell/gene QC filters, biomarker cross-check, batch-effect flags
5	Model development and separation of datasets	4-fold multi-site CV, cross-lineage CV, held-out spike-in validation
6	Independence of training and test data	Leave-site-group-out folds; frozen golden dataset for Q5E comparability
7	Human-AI interface transparency	Probability distribution, per-cell table, QC flags, threshold sensitivity
8	Testing for intended use	Spike-in sweeps, ablation, multi-seed reproducibility, calibration metrics
9	Deployed model monitoring	CalibrationMonitor (KL/JS/Wasserstein/KS) with SQLite persistence
10	Periodic model review and retraining	PCCP with defined retraining triggers and rollback plan

1.4 PCCP — Predetermined Change Control Plan (FDA, Dec 2024)

A PCCP template is maintained (paper/pccp_template.md) describing planned modifications (weights, feature space, thresholds, site priors), the modification protocol (data requirements, retraining procedure, acceptance criteria, verification) and the impact assessment (benefit-risk, post-market monitoring, rollback). A PCCP-delta is emitted on every model retrain and every promotion is recorded in the audit trail with a reference to the delta document.

2. EMA Position

2.1 EMEA/CHMP/410869/2006 — Guideline on Human Cell-Based Medicinal Products

The guideline requires identity, potency, purity, microbiological safety, viability and tumorigenicity testing for cell-based medicinal products. For purity, residual undifferentiated cells must be below a sponsor-justified threshold, with the analytical choice justified. resipsc supports this justification by providing a quantitative, probability-calibrated residual PSC fraction with bootstrap 95% confidence intervals and an orthogonal biomarker cross-check (OCT4/NANOG/SOX2 panel agreement is reported on every run). A Qualified Person release decision remains defensible when the fraction is at or below 0.001% with orthogonal confirmation; resipsc's HOLD disposition maps to the QP consultation path, and QUARANTINE to a failed purity criterion requiring investigation.

2.2 European QP interaction

Every QC report includes the regulatory context section with the EMA rationale, the observed fraction with CI, and recommended orthogonal assays (ddPCR for LIN28A/ESRG, immunostaining for OCT4/TRA-1-60, or histological embryoid-body assay). The signature block reserves an explicit QP approval line, and the audit trail captures who reviewed what and when.

3. ICH Guidelines

3.1 ICH Q5E — Comparability of Biotechnological/Biological Products

Every model promotion requires a comparability study between the previous and new model on the golden dataset: sensitivity delta, specificity delta, decision concordance and Cohen's kappa with acceptance criteria ($|\Delta\text{sens}| \leq 0.02$, $|\Delta\text{spec}| \leq 0.02$, $\kappa \geq 0.90$), plus a Bland-Altman analysis of paired per-cell probabilities. The comparability engine and report generator are implemented in the resipsc package and exposed at POST /api/comparability. Results are persisted to results/comparability_report.json and the audit trail records the promotion.

3.2 ICH Q9(R1) — Quality Risk Management

A formal FMEA covers input data quality, model failure modes, infrastructure and governance. Risk priority numbers (RPN = severity × occurrence × detection) drive the control strategy. Highest-ranked items: FM-02 Batch effects between product runs (RPN 192); FM-01 Low cell count (< 200 cells sequenced) (RPN 180); FM-03 Ambient RNA contamination carries marker (RPN 180).

ID	Component	Failure mode	S	O	D	RPN	Class
FM-02	Input data	Batch effects between product runs	8	6	4	192	HIGH
FM-01	Input data	Low cell count (< 200 cells sequenced)	9	4	5	180	HIGH
FM-03	Input data	Ambient RNA contamination carries marker signal	6	5	6	180	HIGH
FM-05	Model	False-negative: tumorigenic lot released	10	2	7	140	MEDIUM
FM-04	Input data	Doublets simulate PSC gene programmes	5	5	5	125	MEDIUM
FM-08	Model	Out-of-lineage product (unseen germ layer)	7	3	5	105	MEDIUM
FM-07	Model	Silent drift of score distribution	7	4	3	84	MEDIUM
FM-09	Infrastructure	Partial file upload / corrupted h5ad	6	3	4	72	MEDIUM
FM-06	Model	False-positive: conforming lot rejected	4	5	3	60	MEDIUM
FM-11	Governance	Unauthorised model swap without comparability	9	2	3	54	LOW
FM-10	Infrastructure	Audit trail unavailable (DB lock)	8	2	2	32	LOW
FM-12	Governance	Privilege escalation / audit tampering	10	1	2	20	LOW

3.3 ICH Q10 — Pharmaceutical Quality System

Change management: PCCP-delta documents and CAPA workflow for drift; Management review: PQS metrics surfaced on the dashboard (calibration status, batch dispositions); Continual improvement: post-market surveillance through per-batch drift monitoring; Knowledge management: model card, validation data, decision rationale and audit trail retained in the repository structure.

4. GAMP 5 Second Edition — Category 5 Custom Software

Phase	Scope	resipsc deliverable
IQ — Installation Qualification	Version manifests, environment configuration captured in	Dockerfile with pinned dependencies + sbom.json; deterministic build; model card v
OQ — Operational Qualification	Functional tests under stressed conditions: input range,	boundary, QC failure injection, security with memory guards; chunked-upload race-cond
PQ — Performance Qualification	End-to-end runs on representative samples against acce	Spike-in sweeps with quantification calibration; acceptance criteria: sensitivity ≥ 99

Note: the current demonstration build is trained on a hybrid corpus (simulated multi-site data plus a real PBMC negative reference). PQ acceptance criteria are defined for the qualified system on clinical manufacturing data, which requires the wet-lab validation programme described in paper/wet_lab_limitation.md.

5. Current Build Performance Summary

Model	ROC-AUC (multi-site CV)	Sensitivity	Specificity
classical	0.8821 ± 0.0245	0.5471	0.9243
mlp	0.8489 ± 0.0312	0.5387	0.9044
ensemble (resipsc)	0.883 ± 0.0211	0.5488	0.9249

Spike-in validation: limit of detection 0.0100% with quantification calibration (sens 0.6947, fpr 0.0). Wet-lab reference LODs: HEC 0.001%, ddPCR (Kuroda) 0.01%, lncRNA flow 0.0001%, in vivo teratoma 0.1-1%.

6. Cybersecurity and Data Privacy

scRNA-seq profiles are treated as PHI under the HIPAA Security Rule (identifiable even after de-identification). Administrative safeguards: RBAC roles (admin, QC officer, QA, viewer) and training; physical: cloud infrastructure controls inherited from the hosting environment; technical: authentication on all mutating API endpoints, 15-minute session timeout, audit trail on every state change, and hash-chained record integrity. Target certifications: HIPAA + SOC 2 Type II (Year 1), ISO 27001 (Year 2), HITRUST r2 (Year 3).

This package describes a Research Use Only demonstration build. It is not a regulatory submission and the software is not FDA-cleared or CE-marked.